

The DN Budget of the Red Band

How many camera counts the verdict rests on, what one count is worth, and why a filter that doubles them is still the fourth thing to do.

In one line. Rv 's verdict rests on a gap of **15 to 23 camera counts** at 622–627 nm, where one count is worth **6 to 10 Rv**. A blue-attenuating filter would raise that gap by about half — and improve σ_{fill} by **2 %**, because the instrument is only a twentieth of the variance. The finding worth keeping is the other one: **a 1 % error in the reference at 622–627 is worth 5.7 to 9.2 Rv**.

Where this came from. Edwin, 2026-09-06, of two fills that read low on the suite row: *"think the two runs marked by the arrows differ from the according runs of the same day/session by something — check this and try to find the reason"*. Chasing that produced a measurement of the instrument's own repeatability, which turned out to answer a different and larger question.

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1. Where the counts are

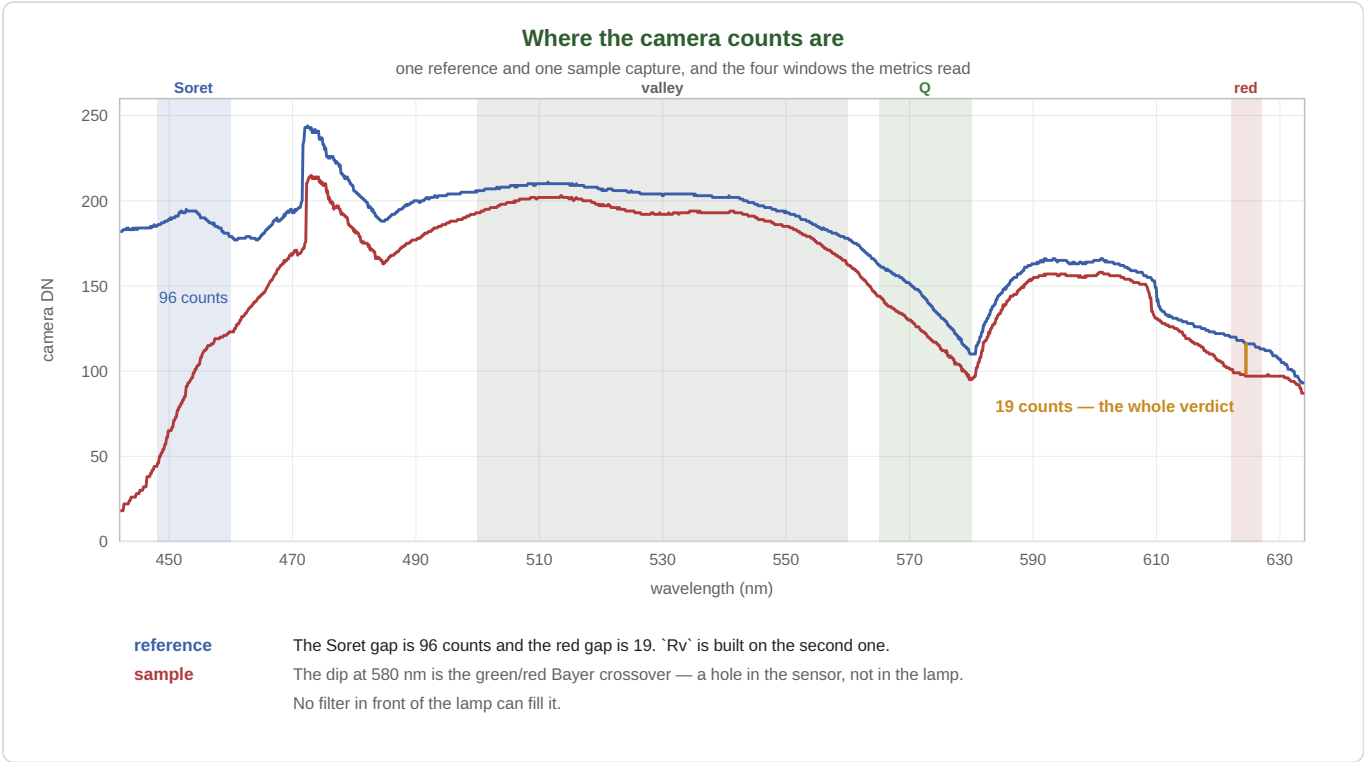


Figure 1. One reference and one sample capture with the four measurement windows marked. The Soret gap is 96 counts; the red gap, on which `Rv` is built, is 19.

The instrument does not measure absorbance. It measures two numbers of camera counts and divides them, and the four windows the metrics read sit at wildly different signal levels. Over the suite's first reads:

oil	red-band gap, reference – sample	1 DN worth
Lugitsch	23.1 counts	8.0 Rv
Ja Natuerlich	20.9 counts	9.6 Rv
Steirerkraft	17.9 counts	8.4 Rv
Spar S-Budget	15.1 counts	5.8 Rv

★ `SPEC_capture_quality.md` §16.40 put it as *"the whole metric lives in 5 to 13 camera counts"*. On the same-jar suite it is 15 to 23, and the brown oil is the poorest — which is `ROADMAP.md` item 7's *"the brown band is 5 DN; it caps brown-oil precision permanently"*, measured.

The sensitivity, for reference. With the `pow2.2` capture decode that every header records,

$$\frac{dRv}{dS} = \frac{100}{A_Q - A_{\text{valley}}} \cdot \frac{2.2 \log_{10} e}{S}$$

⊖ **And the 580 nm dip in Figure 1 is not the lamp.** It is the green/red Bayer crossover — a hole in the sensor. Nothing placed in front of the lamp can fill it, which is §5's ceiling on the whole filter idea.

2. The variance split — the measurement that decides everything below

The suite makes two reads of each fill, and **those two reads share one reference capture and one aliquot**. Their difference therefore contains the instrument and whatever the sample did in the gap, and **nothing of the preparation**. Independent fills of one oil in one sitting contain all of it.

same aliquot, same reference, two reads	median \	ΔRv	1.42 $\Rightarrow$ instrument $\approx$ 1.0 Rv
independent fills, same oil and sitting	pooled sd 4.10 Rv on 13 df		
$\Rightarrow$	the instrument is about 6 % of the variance		

△ The instrument figure is an **over-estimate**, and deliberately so: the 08-28 Steirerkraft pair is left in although its two reads differ by 5–7 Rv, which is the sample browning under the lamp (§16.36), not the instrument. Over-stating the instrument is the safe direction for a document that concludes the instrument is not the problem.

$\Rightarrow$ **Roughly 4.0 Rv of fill-to-fill scatter, of which about 1.0 is the instrument.** Everything else is made before the light reaches the sensor.

3. 🚫 Where an earlier draft of this analysis went wrong

The first version of this arithmetic did not measure the instrument term. It **modelled** it, from the number of distinct integer DN levels present in the 622–627 window — 4 to 9 of them across 36 samples — and arrived at 2.2 to 3.4 Rv. On that basis the filter of §5 was quoted as worth **15 %** on `σ_fill` and **6 %** on the tracker.

The model was too pessimistic by a factor of two to three: 36 samples and 60 averaged frames dither the quantisation far better than a crude `√(distinct levels)` allows. The repeat-read measurement of §2 replaced it, and the filter's value fell to **2 %**.

★ **The lesson is the file's own rule arriving from a new direction:** a number that can be measured must not be modelled. The distinct-level count was a plausible construction with no data under it, and it survived one evening of being quoted before the data that contradicted it was two commands away the whole time.

4. ★★★ What a 1 % reference error costs — the finding worth keeping

Rv reads the red band against the reference captured for that same fill. A reference that is 1 % low at 622–627, relative to its own valley window, moves the answer by:

oil	dRv for a 1 % reference error at 622–627
Ja Natuerlich	9.2 Rv
Steirerkraft	8.1 Rv
Lugitsch	6.5 Rv
Spar S-Budget	5.7 Rv

That is the entire scale of every difference argued over on 2026-09-06 — the Steirerkraft between-night drop (−7.8 Rv), the Ja Natuerlich within-sitting spread (9.9 Rv), both "outlier" fills.

4.1 And it is not hypothetical

Of the two fills that read low on the suite row, 20260906BillaJaNatuerlichD and 20260831SparSBudgetD , the reference legs were compared by **substitution** — recompute each with its session-mates' reference shape, rescaled to its own level:

fill	with its own reference	with the mates' reference shape	difference
Spar S-Budget D	17.81	17.97	+0.16
Ja Natuerlich D	109.05	113.62	+4.57

Spar D's reference is clean and its deviation is in the sample. **Ja Natuerlich D's reference is 0.6 % low in the red band relative to its own valley** — where C and E agree with each other to 0.02 % — and that single 0.6 % accounts for **about 40 % of its 10.9 Rv deviation**.

⇒ ★★★★★ ROADMAP.md item 3 asks for a red reference channel at 622–627 in monitorRecord and prices it at "≈5 Rv, the size of σ_fill itself". This is a measured instance of exactly that, and it was invisible to every field the report records. monitorRecord's columns are qPercent · soret · valley · qBand — there is no red channel, so the one leg that moved could not be seen at the bench, in the report, or in any diagnostic until it was reconstructed from the attached frames.

5. The blue-attenuating filter

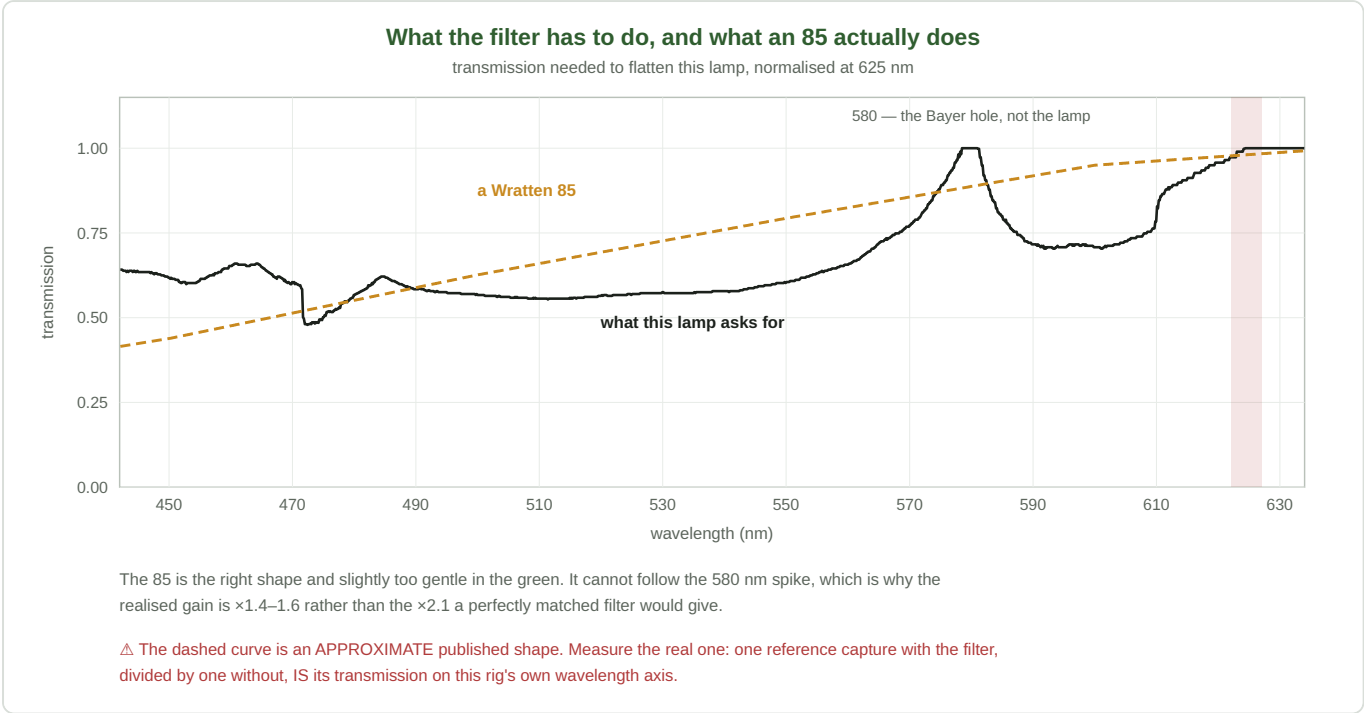


Figure 2. The transmission this lamp asks for, against an approximate Wratten 85. The filter is the right shape and cannot follow the 580 nm spike, which is a sensor artefact.

The idea. The lamp plus the Bayer response is bright in the blue-green and dim in the red — peak-to-red is **2.08** — and the reference peaks within **2–5 %** of the 255 ceiling, so there is no exposure headroom at all. A filter that attenuates the blue lets the exposure rise, and the red gains everything the blue gives up.

★ **It costs nothing in accuracy.** The filter sits in both legs, so it cancels in $T = S/R$ exactly — the same argument as the jar's 0.859 in `DOC_jar_transmission.md` §5.1. It changes the DN distribution and nothing else.

5.1 What it buys, modelled on a Wratten 85

oil	counts now	with an 85	gain	1 DN now	1 DN after
Lugitsch	23.1	35.7	$\times 1.55$	8.0	5.2
Ja Natuerlich	20.9	30.0	$\times 1.43$	9.6	6.7
Steirerkraft	17.9	26.1	$\times 1.46$	8.4	5.8
Spar S-Budget	15.1	21.8	$\times 1.44$	5.8	4.0

The required exposure multiplier comes out at **$\times 1.57$** , against an 85's published filter factor of ≈ 1.5 — an independent check that the model is not nonsense.

5.2 — And what that is worth

$$\sigma_{\text{fill}} : 4.10 \rightarrow 4.03 \quad 2 \%$$

Because the instrument is 6 % of the variance (§2), scaling it by 0.68 moves almost nothing. $\triangle$ For the history tracker, where a between-session term of ≈ 4.8 Rv dominates, it is smaller still.

$\Rightarrow$ **Three fills spread over three evenings is worth roughly twenty times this filter, and costs nothing.**

5.3 What it is still worth having

★ the brown-oil floor	Spar's 15 counts is the poorest signal in the archive and ROADMAP item 7 calls it a permanent cap. 15 $\rightarrow$ 22 counts is a real change to a hard limit, whatever it does to today's σ
★ the DN guard	the Soret sample level moves 44 $\rightarrow$ ≈ 28 DN, into the guard's 20–50 target. Every Steirerkraft fill has been flagged <code>too-dilute</code> ; that stops
★ it is permanent and free thereafter	one screw-on, no recalibration, no metric change, no archive break

5.4 Choosing and mounting one

The type is a colour-conversion "warming" filter — Wratten **85** (not 85B, which costs five more counts off the Soret for no extra red), sold as an ordinary screw-in camera filter. **Glass, not gel:** resin and gel can fluoresce, and fluorescence is the one failure that would corrupt the measurement rather than announce itself.

⊖ Do not buy on the datasheet. The rig measures its own filter: one reference capture with it and one without, and the ratio is `T( $\lambda$ )` on this rig's own wavelength axis. Acceptance: smooth and monotonic through 500–620, ≤ 1.00 everywhere (above 1 means fluorescence), and `T(510)/T(625)` between 0.55 and 0.65.

Mounting at the camera rather than the lamp — which is the only option on the current rig — keeps the whole counts argument intact, because the filter cancels wherever it sits in the beam. It loses one secondary benefit and carries two extra checks:

- **⊖ lost:** at the lamp exit it would also cut the blue dose reaching the sample, and §16.36 measured that light browns the fill. At the camera it does not.
- $\triangle$ **in front of the lens, never between lens and sensor.** A plane-parallel plate in a converging beam shifts the focal plane by about a third of its thickness, which would defocus the spectrum and broaden every band.
- $\triangle$ **check the wavelength axis.** Adding glass can displace the spectral image; cross-correlate a reference with and without, as `diagnostics/d2r_all_runs.py` does for fills. Uncoated glass close to the optics can also ripple; look for that in the measured `T( $\lambda$ )`.

The test, one evening. One fill, two captures — filter off at the pinned exposure, filter on at $\approx 1.6\times$. Three things must hold: `A( $\lambda$ )` agrees within σ_{fill} across the window, the wavelength axis has not moved, and the red-band gap has grown. $\triangle$ `DevSpectralPlugin` pins the exposure at 90 (§16.39.5); that constant changes with the filter.

6. What this document claims, ranked

★★ measured	the red band carries 15–23 counts and one count is worth 5.8–9.6 Rv
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★★ measured	the instrument is ≈ 1.0 Rv of a 4.10 Rv fill-to-fill scatter — about 6 %
★★ measured	a 1 % reference error at 622–627 is worth 5.7–9.2 Rv, and one fill on 2026-09-06 carried 0.6 % of it
★ modelled	a Wratten 85 would raise the red gap by $\times 1.4$ – 1.6 and σ_{fill} by 2 %. The filter's own curve is assumed, not measured
⊖ retracted	the 15 % figure of §3, and with it the claim that quantisation is a major term
⊖ not claimed	that more counts would fix any invariance observed on 2026-09-06. They would not; §2 is why